

# Stock Market Trend Prediction Model using Deep Learning based Sentiment Analysis of Financial Data

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**Abstract**—Creating a forecasting model to identify the trend in the stock market is difficult due to the complex nature and behavior of investors participating in the stock market. The existing models and studies have mainly used past prices and other technical indicators based on past prices or volume to predict future prices or trends instead of Behavioral finance for the Indian market. This study demonstrates a novel neural network-based stock market trend prediction model, which employs sentiment analysis data of published financial information. Deep learning-based sentiment analysis applications are used to identify investor sentiment based on published financial information on Macroeconomic and Microeconomic factors. The deep learning-based trend prediction model using sentiment analysis data has a more than 96% accuracy rate. The study also provides further evidence of behavioral finance theory of herd mentality in the Indian market. The result of research will benefit in the field of financial investment and risk management in the Indian stock market.

**Keywords**—sentiment analysis, stock market, deep learning, forecasting.

## I. INTRODUCTION

A diverse range of investors utilize financial instruments such as stocks and derivatives to make both short- and long-term investments, making the financial markets a form of investment institution that attracts a large number of people. The value of stocks and derivatives is influenced by a multitude of factors, most notably features of the economy, both big and minor. Investors use a range of strategies to safeguard their investments and raise the possibility of a profit. The Indian stock market has recently seen a large influx of capital [1]. Even if they aim to gain money, market volatility causes many individual investors to lose money. The stock market is influenced by a multitude of economic variables, including as shifts in the value of western markets, fluctuations in the price of crude oil, the pace at which wars escalate, the actions of central banks, and investor sentiment [2]. These economic benchmarks are providing guidance to emerging markets like India's. As per the Behavioural finance theory and Adaptive market hypothesis, the investor behaviour is a key factor affecting the volatility[3]. With the development of cutting-edge AI technologies, research has gained pace to understand the patterns of stock price movement and volatility. However in these research a key aspect to understand the behaviour of investors using AI technologies is missing[4].

In this article the Deep learning based Sentiment analysis services is used to understand the sentiment of published Macro and Micro economic information, like the investors do the analysis based on these information[5]. This study

identifies the key macroeconomic information such as volatility of global stock exchanges, war escalations, GDP data, Government policies, Inflation information, Interest rate details, Central bank actions and microeconomic information such as company earning, new product launch, AGM related information through Deep learning based NLU (Natural Language Understanding) models[6]. Based on the result of these models a deep learning based neural network model is created to predict the trend of Indian stock market. In this research, the NIFTY 50 Index listed companies are used because these list of companies are comprehensive representation of the Indian stock market. The total market capitalization of these companies amounts to \$2.23 trillion[1].

Here are the goals of this research:

- To extract investor sentiment from published financial information using deep learning based NLU applications.
- To establish the relationship between stock trend and sentiment analysis data of macro and micro economic information.
- To create a deep neural network model to forecast the trend of stocks in Indian stock market using sentiment analysis data.
- To use various statistical test to compare the predicted result against actual result.

In the following sections of this paper, explains about the related works in the same field, proposed methodology and test results.

## II. RELATED WORK

Behavioural finance questions the presumptions of conventional financial theories such Efficient Market Hypothesis (EMH), which hold that markets are efficient and investors are logical. Rather, behavioural finance holds that investing decisions are largely influenced by psychological prejudices and emotions. Important behavioural finance biases include in:

Overconfidence Bias: Investors overstate their capacity to forecast changes in the market, which results in too risky behaviour [3]. Herding Behaviour: Usually following the herd, investors might cause market booms or collapses [7]. Loss Aversion: Investors make risk-averse judgments because they experience the more profound suffering of losses than the gratification of winnings [8]. Studies conducted in the framework of the Indian stock market have revealed that behavioural biases in investor attitude greatly affect market patterns [9].

Andrew Lo's 2004 introduction of the Adaptive Market Hypothesis suggests that market efficiency is not fixed but changes over time, therefore reconciling the contradicting ideas of EMH and behavioural finance. AMH claims that the environment and investor conduct may cause markets to show both efficiency and inefficiency [10]. Studies shown that the Indian marketplaces display different degrees of efficiency; certain time periods show indications of predictability because of shifting market circumstances[11]. Research done by Baker & Wurgler shows how investor emotion affects the fluctuation in stock price. The Indian stock market is affected by the volatility and returns of international stock markets [12]. Recent studies imply that stock market volatility and returns are influenced by public view of news events [13]. Sentiment analysis on textual data helps one to grasp the attitude on an investor. Natural language processing (NLP) method sentiment analysis finds the sentiment conveyed in textual data including financial reports, social media, and news items. In finance, it is used to determine the emotional tone of the

market in order to forecast next price moves. Important element of market activity, investor sentiment may be captured via sentiment analysis. Sentiment analysis has been widely applied to predict stock price movements in India. Studies like Jain et al. found that news sentiment extracted from financial articles could be used as a leading indicator of stock price movements [14]. Similarly, studies used sentiment analysis on social media data (e.g., Twitter) to predict intraday stock market movements in India[15]. Predicting stock market trends is a complex task due to the myriad factors influencing price movements. Traditional methods rely on technical analysis, fundamental analysis, and econometric models. However, with advancements in artificial intelligence, new approaches such as sentiment analysis and machine learning have gained traction. With the advent of machine learning ideas, scientists set out to create prediction models that would make use of these methods [16]. Table 1 shows recent studies for predicting stock market using past price information.

TABLE I. THE SCHOLARLY ARTICLES STUDIED FOR THEIR RELEVANCE TO STOCK MARKET FORECASTING

| Reference | Market        | Dataset used        | Algorithm Used                          | Result / Discussion                   | Year |
|-----------|---------------|---------------------|-----------------------------------------|---------------------------------------|------|
| [17]      | Indian Market | NIFTY 50 index      | CNN                                     | Accuracy of 75.41%                    | 2020 |
| [18]      | Indian Market | NIFTY 50 index      | LSTM                                    | Accuracy through R2 0.8388            | 2022 |
| [19]      | Indian Market | BSE & NSE IT Index  | LSTM                                    | MAPE of 1.03%                         | 2022 |
| [20]      | Indian Market | Indian Stock Market | LSTM, Random Forest                     | MAPE of 1.36% for LSTM                | 2022 |
| [21]      | Indian Market | NIFTY 50 index      | Decision Tree, Random Forest, SVM, LSTM | Highest Accuracy of 80.44% using LSTM | 2023 |
| [22]      | Indian Market | NIFTY 50 index      | Chaos, LSTM, &PR                        | MAPE of 1.21                          | 2023 |

### III. PROPOSED METHOD

The research work involves extraction of necessary financial and stock market data, processing the data with proper format, sentiment analysis of financial information

using various deep learning based NLU applications, fusion of stock market trend data with sentiment analysis data, creation of trend forecasting model using deep neural network and evaluating the results. Figure 1 depicts the process flow diagram at a high level.

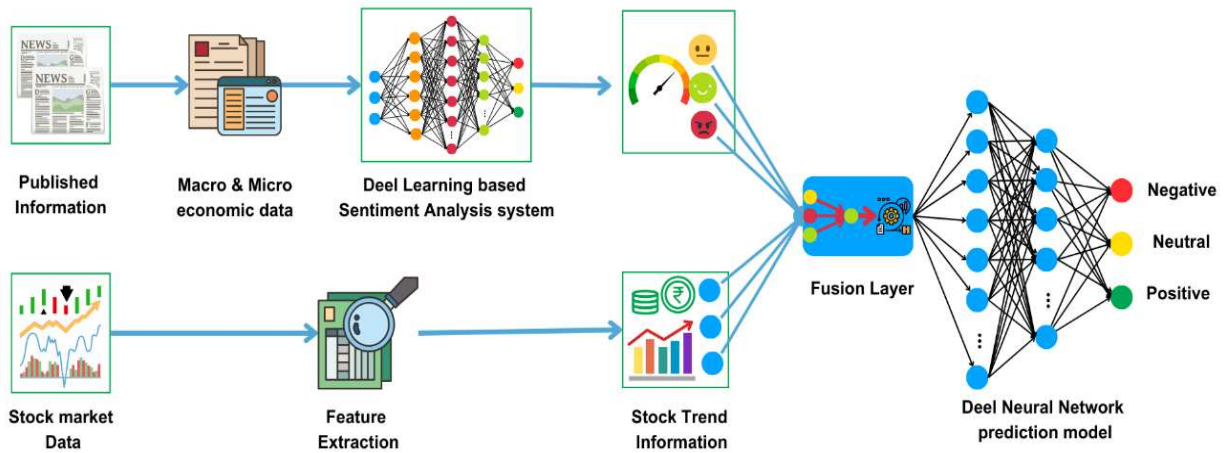


Fig. 1. High level process flow diagram illustrating data extraction, feature selection, Deep learning based sentiment analysis process, and Deep neural network prediction model

#### A. Data Extraction

The input data has collected from November 29, 2021, to December 29, 2023 period. This time frame is selected because of lower volatility after Covid-19 pandemic impact. The majority of the data is taken from two highest viewed Indian newspapers websites, namely The Economic Times and Hindustan Times. The financial data is filtered for India specific Macroeconomic and Microeconomic factors [23]. Similarly the stock price data is taken from NSE website for NIFTY 50 index constituent companies which are top 50

companies of India. One day return information is derived from the price using (1).

$$R_{S,t} = \ln\left(\frac{P_{S,t}}{P_{S,t-1}}\right) \quad (1)$$

Where  $R_{S,t}$  is the return of stock S at time t,  $P_{S,t}$  is the price of stock S at time t and  $P_{S,t-1}$  is the price of the stock S at time t-1.

### B. Sentiment Analysis Process

Deep learning based Natural Language Understanding (NLU) application is utilized to do sentiment analysis on the

pre-processed economic information. Figure 2 represents the high level architecture.

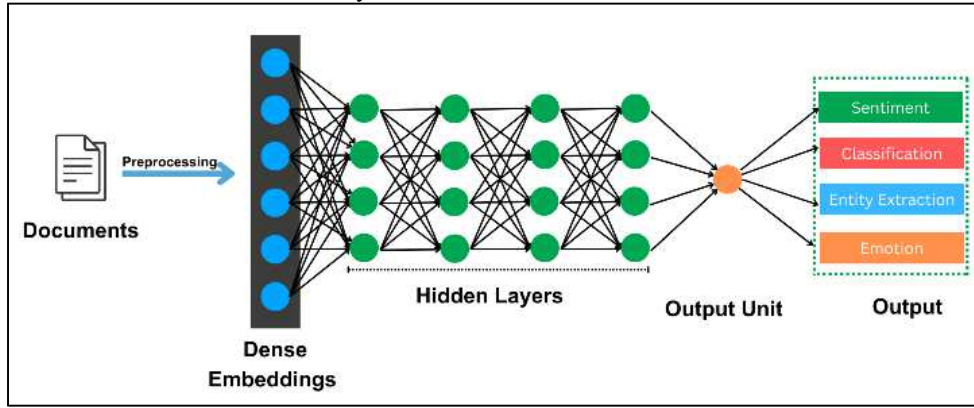


Fig. 2. High level process flow diagram illustrating Deep learning based sentiment analysis process

The unique NLU model is utilized to process each economic input data and derive the sentiment analysis score. The sentiment score runs from -1 to +1, and the label assigned to the data can be either "Positive", "Negative", or "Neutral" depending on the sentiment score. If  $Macro_t$  is derived from textual sentiment analysis at time  $t$  and  $\theta$  is a small threshold representing neutral sentiment then  $S_{Macro}$  be defined in equation (2),

$$S_{Macro} = \begin{cases} \min(1, Macro_t), Positive, & \text{if } Macro_t > 0 \\ \max(-1, Macro_t), Negative, & \text{if } Macro_t < 0 \\ 0, Neutral, & \text{if } |Macro_t| \leq \theta \end{cases} \quad (2)$$

If  $Micro_t$  is derived from textual sentiment analysis at time  $t$  and  $\theta$  is a small threshold representing neutral sentiment then  $S_{Micro}$  can be represented in equation (3),

$$S_{Micro} = \begin{cases} \min(1, Micro_t), Positive, & \text{if } Micro_t > 0 \\ \max(-1, Micro_t), Negative, & \text{if } Micro_t < 0 \\ 0, Neutral, & \text{if } |Micro_t| \leq \theta \end{cases} \quad (3)$$

### C. Deep Neural Network based prediction model

The relation between trend of the stock and sentiment of macro and Micro economic data can be defined as regression model in equation (4),

$$R_{Trend,t} = \alpha + \beta_1 S_{Macro} + \beta_2 S_{Micro} + \varepsilon \quad (4)$$

Where  $\alpha$  is the intercept,  $\beta_1$  is the coefficient that measures the sensitivity of stock returns to macroeconomic sentiment,  $\beta_2$  is the coefficient that measures the sensitivity of stock returns to microeconomic sentiment and  $\varepsilon$  is the error term. To create a neural network model that captures the relationship between trend and sentiment (derived from both macro and microeconomic data), the model can be expressed as a series of equations (5) involving input layers, hidden layers, activation functions, and output layers

$$f_{DLN}(S_{Macro,t}, S_{Micro,t}, S_{Macro,(t-1)}, S_{Micro,(t-1)}, \dots) = \hat{y}_{t+1} \quad (5)$$

1) *Input Layer*: The input consists of macro and microeconomic sentiment scores at time  $t$ . Defining the input vector  $x_t$  as equation in (6),

$$x_t = \begin{cases} S_{Macro,t} \\ S_{Micro,t} \end{cases} \quad (6)$$

2) *Hidden Layers*: These layers apply weights and activation functions to model complex relationships between inputs and outputs. For the multiple hidden layer, the output from the previous hidden layer becomes the input to the next hidden layer. For the  $m$  hidden layer the equation can be (7):

$$h_k^{(m)} = f(w_k^{(m)} \cdot h^{(m-1)} + b_k^{(m)}) \quad (7)$$

Where,  $w_k^{(m)}$  is the weight vector for neuron  $k$  in the  $m$  hidden layer,  $b_k^{(m)}$  is the bias for neuron  $k$  in the  $m$  hidden layer and  $f$ : Activation function  $f(z) = \max(0, z)$

3) *Output Layer*: The output is the predicted return based on the input sentiment scores. The final output is a linear combination of the activations from the last hidden layer. Let  $\hat{y}_{t+1}$  represent the predicted trend at time  $t + 1$  equation (8).

$$\hat{y}_{t+1} = g(w^{(o)} \cdot h^{(m)} + b^{(o)}) \quad (8)$$

Where  $w^{(o)}$  is the weights of the output layer,  $b^{(o)}$  is the bias of the output layer,  $h^{(m)}$  is the activations from the last hidden layer (if there are  $m$  hidden layers)  $g$ : The linear activation function ( $g(z) = z$ ) in the output layer

4) *Training the neural network*: The model is trained to minimize a loss function mean squared error for regression, the loss function can be defined equation (9),

$$\mathcal{L}(\hat{y}_{t+1}, y_{t+1}) = \frac{1}{N} \sum_{t=1}^N (\hat{y}_{t+1} - y_{t+1})^2 \quad (9)$$

Where,  $y_{t+1}$  is the actual Trend at time  $t + 1$ ,  $\hat{y}_{t+1}$  is the predicted Trend at time  $t + 1$ , and  $N$  is the number of training samples. Chi-square test is performed to understand the association between 1-day return trend and the independent variables (macro and micro sentiment). The deep learning model performance on the predicted result for the one day

trend is evaluated based on the area under the curve for sensitivity and specificity curve for the actual data. The equation (10) for AUC can be,

$$AUC = \int_0^1 \frac{TP}{TP+FN} d \frac{FP}{FP+TN} \quad (10)$$

Where TP is true positive, TN is True Negative, FP is False positive and FN is False Negative.

#### IV. RESULTS AND DISCUSSION

##### A. Stock price one day return data

The stock price data is taken from NSE website for NIFTY 50 index constituent companies. One day return data is derived from the closing prices based on (1). One day return data is used, to keep it time independent. Table 2 shows the result of Autocorrelation and Box-Ljung test and Table 3 shows the test result of Run test to understand the time dependency on the one day return data.. The test results show the Autocorrelation value is less than 0.05 for all values till 10th lag which signifies the data series for one day return is time independent. Similarly the Run test result shows the test value is 0.11113 with asymptomatic significance 0.339, which is greater than 0.05. Hence the data series comprising one day return for 50 stocks is time independent.

TABLE II. AUTOCORRELATION TEST AND BOX-LJUNG TEST RESULTS

| Lag | Autocorrelation | Std. Error | Box-Ljung Statistic |    |       |
|-----|-----------------|------------|---------------------|----|-------|
|     |                 |            | Value               | df | Sig.  |
| 1   | 0.004           | 0.013      | 30.719              | 1  | 0.000 |
| 2   | -0.014          | 0.013      | 31.755              | 2  | 0.000 |
| 3   | 0.005           | 0.013      | 31.899              | 3  | 0.000 |
| 4   | -0.003          | 0.013      | 31.945              | 4  | 0.000 |
| 5   | -0.037          | 0.013      | 39.613              | 5  | 0.000 |
| 6   | 0.001           | 0.013      | 39.618              | 6  | 0.000 |
| 7   | -0.004          | 0.013      | 39.705              | 7  | 0.000 |
| 8   | 0.005           | 0.013      | 39.831              | 8  | 0.000 |
| 9   | 0.023           | 0.013      | 42.787              | 9  | 0.000 |
| 10  | 0.004           | 0.013      | 42.870              | 10 | 0.000 |

TABLE III. RUN TEST RESULTS

| Test Value             | 0.111134631745884 |
|------------------------|-------------------|
| Cases < Test Value     | 2834              |
| Cases >= Test Value    | 2834              |
| Total Cases            | 5668              |
| Number of Runs         | 2799              |
| Z                      | -0.956            |
| Asymp. Sig. (2-tailed) | 0.339             |

##### B. Sentiment Analysis using Deep learning NLU services

TABLE IV. SENTIMENT ANALYSIS RESULT FOR FINANCIAL DATA

| Platform            | Output               | Accuracy |
|---------------------|----------------------|----------|
| IBM Watson          | Score between [-1,1] | 97%      |
| Microsoft Azure NLU | Score between [0,1]  | 94%      |
| Google NLU          | Score between [-1,1] | 90%      |
| Open AI             | Score between [-1,1] | 94%      |
| finBERT             | Score between [0,1]  | 93%      |

For the sentiment analysis of macro and micro economic data, multiple Deep learning based sentiment analysis systems were used. The result with a sample of data is compared against IBM Watson NLU, Google Cloud NLU, Microsoft Azure NLU, OpenAI NLU and FinBERT systems. Clearly IBM Watson's accuracy is better than other services [13].

Table 4 shows the accuracy of each NLU based sentiment analysis system that were considered.

Figure 3 presents the programming procedures performed for the sentiment analysis from both Macroeconomic and Microeconomic data.

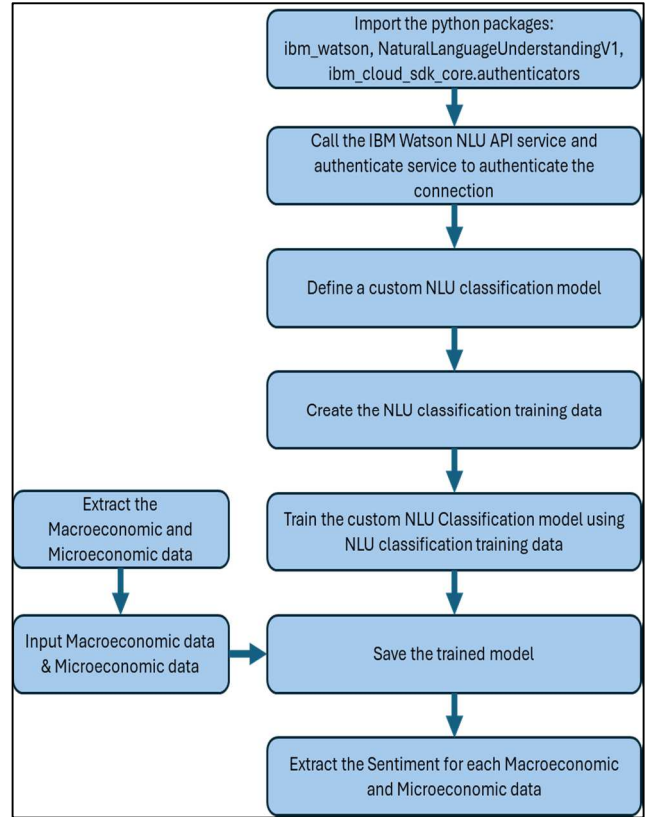


Fig. 3. High level program structure to extract sentiment from Macroeconomic data and Microeconomic data

##### C. Trend prediction model using Deep Neural Network

IBM SPSS tool has been used to create the Deep Neural network. Table 5 represent the construction of Deep Neural Network. The training set is 81.1% data and testing set is 18.9% data.

TABLE V. CONSTRUCTION OF DEEP NEURAL NETWORK MODEL USING SPSS

| Input Layer     | Factors                           |   | 1             | Macro Sentiment    |
|-----------------|-----------------------------------|---|---------------|--------------------|
|                 |                                   |   | 2             | Micro Sentiment    |
|                 | Covariates                        |   | 1             | Macro Score        |
|                 |                                   |   | 2             | Micro Score        |
|                 | Number of Units                   |   |               | 8                  |
|                 | Rescaling Method for Covariates   |   |               | Standardized       |
| Hidden Layer(s) | Number of Hidden Layers           |   |               | 2                  |
|                 | Number of Units in Hidden Layer 1 |   |               | 5                  |
|                 | Number of Units in Hidden Layer 2 |   |               | 3                  |
|                 | Activation Function               |   |               | Hyperbolic tangent |
|                 |                                   |   |               |                    |
| Output Layer    | Dependent Variables               | 1 | 1 Day Trend   |                    |
|                 | Number of Units                   |   | 3             |                    |
|                 | Activation Function               |   | Softmax       |                    |
|                 | Error Function                    |   | Cross-entropy |                    |

Figure 4 depicts the deep neural network for the prediction of Trend using sentiment analysis data from Macro and Micro economic financial information.

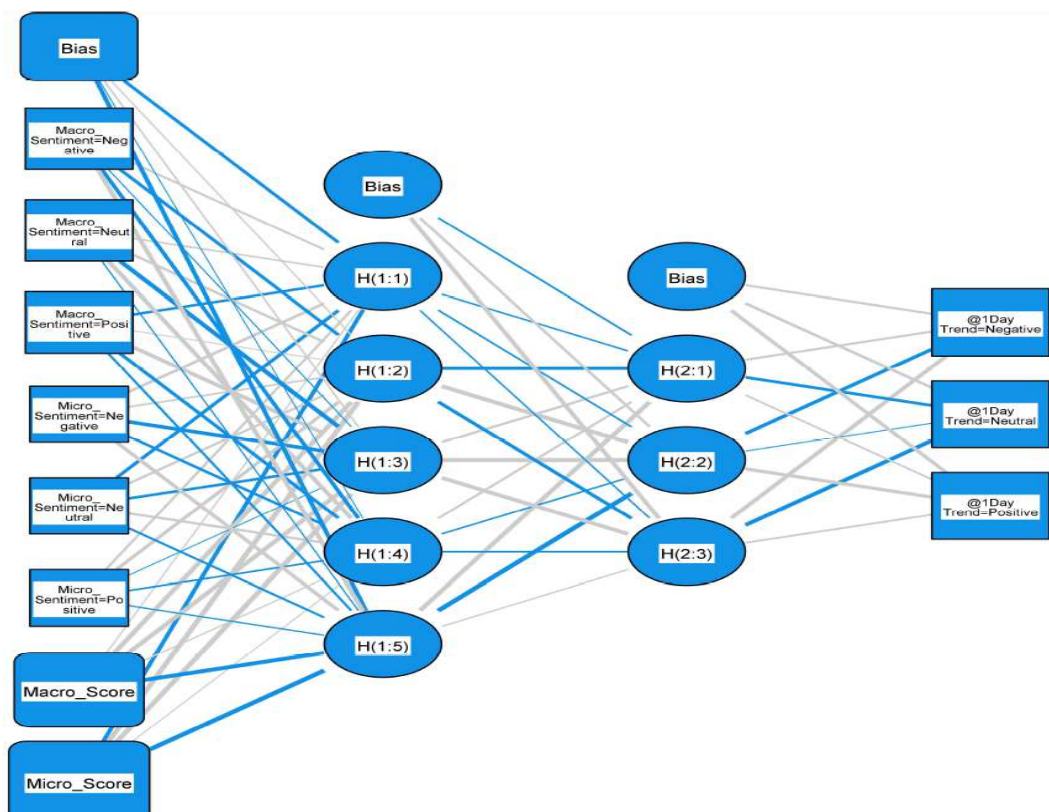


Fig. 4. Trend prediction model using Deep Neural Network

The descriptive statistics of the input dataset is represented in Table 6. Total 5668 observation for all the NIFTY 50 index stocks are collected.

The Chi-square result suggests that there is a statistically significant relationship between the dependent variable (1-day

return trend) and the independent variables (macro and micro sentiment). The large Chi-Square value further supports that this association is quite strong, meaning the sentiment variables (macro and micro) likely have a meaningful impact on the 1-day return trend. Table 7 shows the Chi-square test result.

TABLE VI. SUMMARY OF DESCRIPTIVE STATISTICS

|                       |                   | 1day Return | Sentiment Score | Macro Score | Micro Score |
|-----------------------|-------------------|-------------|-----------------|-------------|-------------|
| <b>N</b>              | <i>Statistic</i>  | 5668        | 5668            | 5668        | 5668        |
| <b>Mean</b>           | <i>Statistic</i>  | 0.158951    | 0.022934        | -0.010328   | 0.055852    |
|                       | <i>Std. Error</i> | 0.030706    | 0.004760        | 0.005535    | 0.008064    |
| <b>Std. Deviation</b> | <i>Statistic</i>  | 2.311768    | 0.358385        | 0.416733    | 0.607099    |
|                       | <i>Std. Error</i> | 0.033       | 0.033           | 0.033       | 0.033       |
| <b>Variance</b>       | <i>Statistic</i>  | 5.344       | 0.128           | 0.174       | 0.369       |
| <b>Skewness</b>       | <i>Statistic</i>  | -0.617      | -0.146          | -0.100      | -0.168      |
|                       | <i>Std. Error</i> | 0.033       | 0.033           | 0.033       | 0.033       |
| <b>Kurtosis</b>       | <i>Statistic</i>  | 17.029      | -0.711          | -0.838      | -1.474      |
|                       | <i>Std. Error</i> | 0.065       | 0.065           | 0.065       | 0.065       |

TABLE VII. CHI-SQUARE TEST RESULT

|                     | Value    | df | Asymptotic Significance (2-sided) |
|---------------------|----------|----|-----------------------------------|
| Pearson Chi-Square  | 5616.008 | 4  | 0                                 |
| Likelihood Ratio    | 975.629  | 4  | 0                                 |
| McNemar-Bowker Test | 6.111    | 3  | 0.106                             |
| N of Valid Cases    | 5668     |    |                                   |

The deep neural network model created on IBM SPSS software executed with the input data. Both the training and testing accuracy is same with incorrect predictions only 3.9%. The summary of the Deep Neural model is represented in Table 8.

One day trend is predicted based on training and testing data. For the testing data, the model achieved overall 96.1%

accuracy for predicting one day trend. The model can predict positive trends 97.7% compared to negative trends for 94.3% and neutral trends for 92.3%. Classification result from the deep neural network model is showing in Table 9.

TABLE VIII. SUMMARY OF DEEP NEURAL NETWORK MODEL PERFORMANCE.

|                 |                                      |                                                 |
|-----------------|--------------------------------------|-------------------------------------------------|
| <b>Training</b> | <i>Cross Entropy Error</i>           | 738.425                                         |
|                 | <i>Percent Incorrect Predictions</i> | 3.9%                                            |
|                 | <i>Stopping Rule Used</i>            | 1 consecutive step(s) with no decrease in error |
| <b>Testing</b>  | <i>Training Time</i>                 | 0:00:00.32                                      |
|                 | <i>Cross Entropy Error</i>           | 183.257                                         |
|                 | <i>Percent Incorrect Predictions</i> | 3.9%                                            |

Figure 5 represents the pseudo-probabilities against the dependant variable.

TABLE IX. CLASSIFICATION RESULT FROM DEEP NEURAL NETWORK MODEL.

| Sample   | Observed        | Predicted |         |          |                 |
|----------|-----------------|-----------|---------|----------|-----------------|
|          |                 | Negative  | Neutral | Positive | Percent Correct |
| Training | Negative        | 1982      | 0       | 114      | 94.6%           |
|          | Neutral         | 0         | 56      | 4        | 93.3%           |
|          | Positive        | 58        | 0       | 2300     | 97.5%           |
|          | Overall Percent | 45.2%     | 1.2%    | 53.6%    | 96.1%           |
| Testing  | Negative        | 493       | 0       | 30       | 94.3%           |
|          | Neutral         | 1         | 12      | 0        | 92.3%           |
|          | Positive        | 14        | 0       | 604      | 97.7%           |
|          | Overall Percent | 44.0%     | 1.0%    | 54.9%    | 96.1%           |

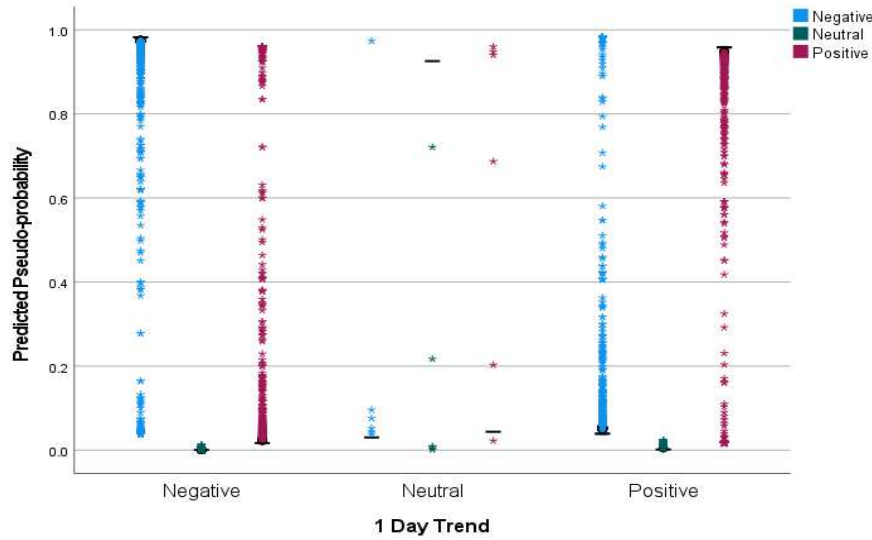


Fig. 5. Predicted psudo probabilities against the one day trend.

Figure 6 represents the sensitivity and specificity for the predicted one day return classifications. Area under the curve for the sensitivity and specificity pair is presented in Table 10. The results show 0.968 for both positive and negative prediction and 0.986 for neutral prediction.

TABLE X. AREA UNDER THE CURVE (AUC) FOR SENSITIVITY - SPECIFICITY CURVE

| 1 Day Trend | Area     |       |
|-------------|----------|-------|
|             | Negative | 0.968 |
|             | Neutral  | 0.986 |
|             | Positive | 0.968 |

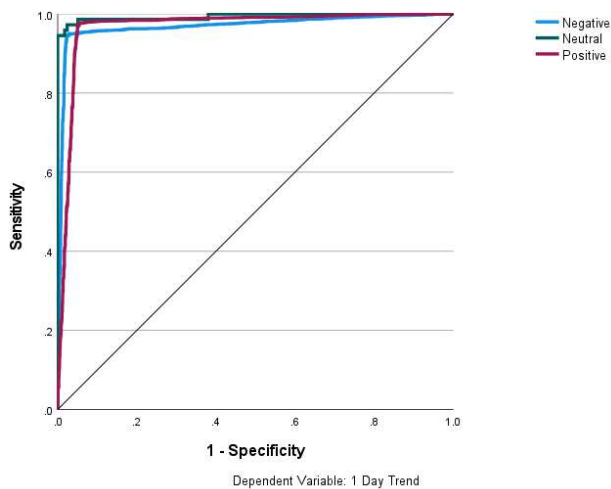


Fig. 6. Sensitivity and specificity curve for the predicted one day return classifications.

## V. CONCLUSION

This study aimed to determine the sentiment of economic information by employing deep learning based NLU techniques. By analyzing the sentiment scores, a correlation between the trends of one day return in the NIFTY 50 index companies with the sentiment of macro and micro economic data is established. In addition, the one day trend of companies listed in NIFTY 50 index is attempted to forecast by utilizing the sentiment data. The experiment data reveals that there is a statistically significant relationship between 1-day return trend and the sentiment from macro and micro economic data. The large Chi-Square value further supports that this association is quite strong, meaning the sentiment variables (macro and micro data) likely have a meaningful impact on the 1-day return trend. Deep neural network (MPL) model is utilized to predict the one day trend using the sentiment analysis data. Our model has achieved overall 96.1% accuracy for predicting one day trend. The model can predict positive trends 97.7% compared to negative trends for 94.3% and neutral trends for 92.3%. The results show the potential use of sentiment analysis in the Indian Financial market to identify the trend of the market based on the macro and micro economic information. The research aims to assist investors who are currently investing or planning to invest in the Indian stock market, specifically in the NIFTY 50 listed stocks or the respective index through its derivative segment or Index funds. By utilizing the predictions, investors will gain insights into the anticipated trend in the market. By analyzing the sentiment score, individuals may accurately predict market trends and make financial decisions appropriately. Based on the result they can readjust their long positions or hold the positions for a better return. The study has opened the scope

for further research to comprehend the analogous investor or consumer opinion regarding the price volatility in alternative marketplaces or regions.

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